

TECHNICAL DESIGN REPORT
POWER ELECTRONICS AND CONTROL SYSTEMS

Flying Capacitor Multilevel: single phase inverter design, control and simulation.

Six-level half-bridge flying capacitor inverter for a medium voltage DC link.

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Revision:

00

Date:

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Abstract: This report documents the design of a single phase inverter built around a six-level half-bridge flying capacitor stage, developed together with Alintel srl for a DC link in the range 3 kV to 5 kV. The converter layout is described first: the flying capacitor banks, obtained as series-parallel arrangements of a base capacitor element, the high-frequency capacitors placed at each level to reduce the effects of the high di/dt , the output LRC filter and the pre-charge circuit, whose charging sequence is driven by a negative ramp of the modulation index. The stray inductances of components and wiring are then included in a high-frequency model of the complete stack, which is used to assess the electrical layout through switching level simulations at 5 kV and 3 kV of input voltage. The second part of the document covers the control: the quadrature components of the output voltage and of the output current are reconstructed by two state observers based on a resonant plant at the fundamental frequency and implemented as Kalman filters; they feed an outer voltage magnitude loop closed around an inner vector current loop. The interleaved modulation among the levels provides the natural balancing of the flying capacitor voltages. Closed loop simulation results cover the pre-charge transient, the no-load operation and the connection of a step load.

REVISION HISTORY

Rev.	Date	Description	Authors
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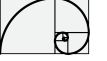
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Nomenclature

Here the list of variables and parameters used along the document:

- C_{fc}^{base} [F]: base capacitance for flying capacitors;
- R_{Cfc}^{base} [Ω]: base resistance of the base module for flying capacitors;
- L_{Cfc}^{base} [H]: base inductance of the base module for flying capacitors;
- C_{hf}^{base} [F]: base capacitance for high frequency capacitors;
- R_{Chf}^{base} [Ω]: base resistance of the base module for high frequency capacitors;
- L_{Chf}^{base} [H]: base inductance of the base module for high frequency capacitors;
- C_{fc}^n [F]: equivalent capacitance of the n-level flying capacitor;
- R_{Cfc}^n [Ω]: equivalent resistance of the n-level flying capacitor;
- L_{Cfc}^n [H]: equivalent inductance of the n-level flying capacitor;
- C_{hf}^n [F]: equivalent capacitance of the n-level high frequency capacitors;
- R_{Chf}^n [Ω]: equivalent resistor of the n-level high frequency capacitors;
- L_{Chf}^n [H]: equivalent inductance of the n-level high frequency capacitors;
- u_{in} [V]: input voltage;
- u_s [V]: half-bridge output voltage;
- u_{out} [V]: output voltage (output voltage of the LRC filter);
- i_{in} [A]: input current;
- i_s [A]: half-bridge output current;
- i_{out} [A]: output current (output current of the LRC filter);
- L_{Fu} [H]: output filter inductance;
- R_{Fu} [Ω]: output filter damping resistor;
- C_{Fu} [F]: output filter capacitance;



1 Converter Layout Description

Figure 1, Figure 3, and Figure 4 show the whole layout, the output filter, and the six-level half bridge flying capacitor respectively, in particular

- Figure 1 shows the pre-charge system;
- Figure 2 shows the electrical layout of the six-level half bridge flying capacitor.
- Figure 3 shows the output LRC filter;
- Figure 4 shows a full model of the six-level half bridge flying capacitor, where high frequency representation of stray parts are included. Configuration of the flying capacitor is also reported;

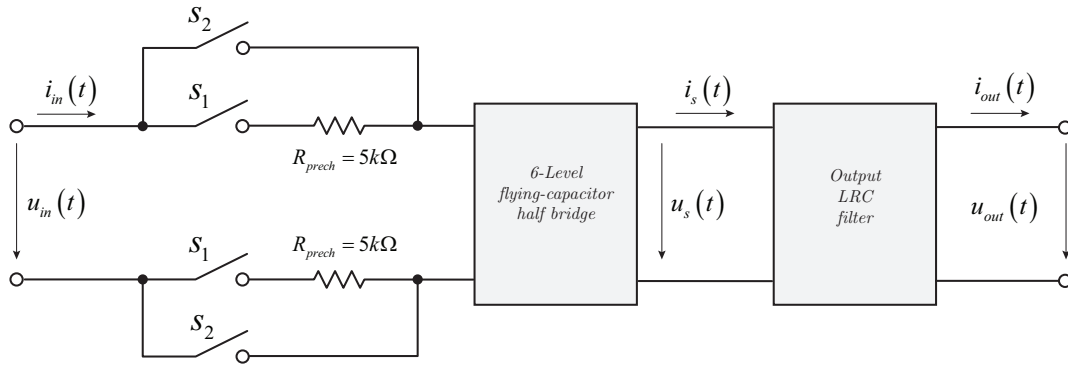


Figure 1: Auxiliary power supply.

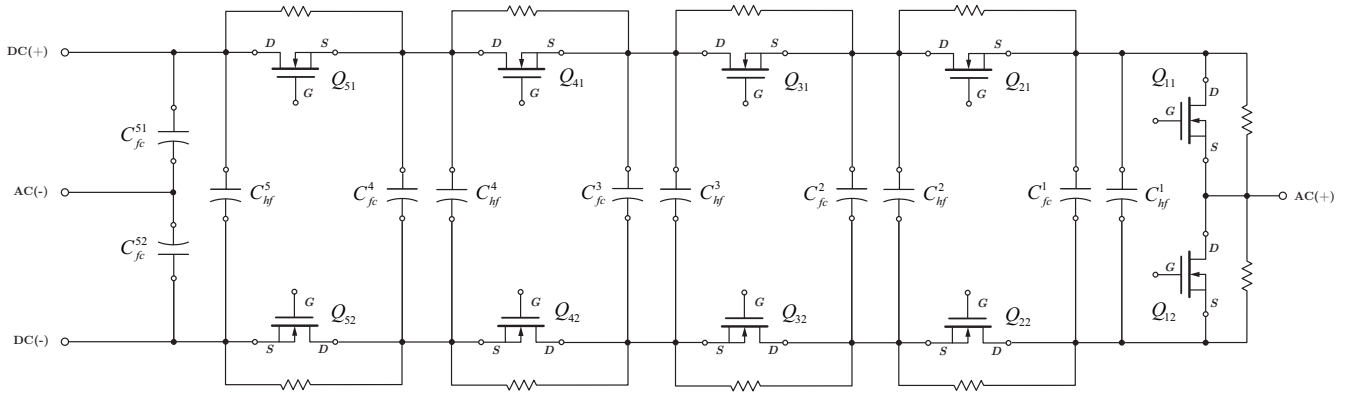


Figure 2: Half-bridge six-level flying capacitor converter.

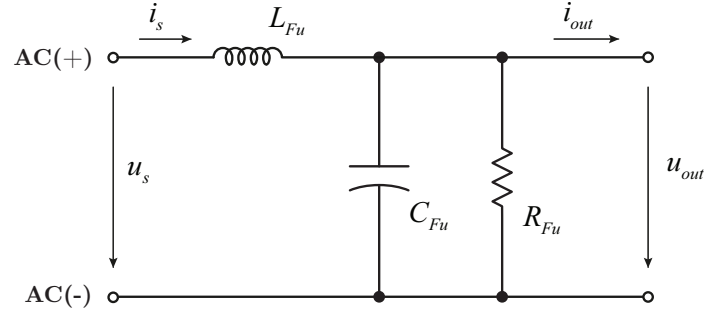
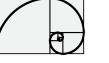


Figure 3: Output filter electrical layout.

According to Figure 4 each flying capacitor bank is created by a series-parallel connection of a *base* capacitor element, e.g. the fourth level of flying capacitor is created by four parallel harm made by for series capacitor connection ($4s$, $4p$) which results as follows

$$\begin{cases} C_{fc}^n = C_{fc}^{base} \\ R_{Cfc}^n = R_{Cfc}^{base} \\ L_{Cfc}^n = L_{Cfc}^{base} \end{cases} \quad (1.1)$$

Different approach is used for the *high-frequency* capacitors* where the connection is made only if series mode. The use of the *high-frequency* capacitor dclinks is made to reduce the effects of the high di/dt .

$$\begin{cases} C_{hf}^n = \frac{1}{n} \cdot C_{hf}^{base} \\ R_{Chf}^n = n \cdot R_{Chf}^{base} \\ L_{Chf}^n = n \cdot L_{Chf}^{base} \end{cases} \quad (1.2)$$

*The electrical layout/connection of the high-frequency capacitors to the corresponding mosfets level is made in a way to minimize the equivalent stray inductance.

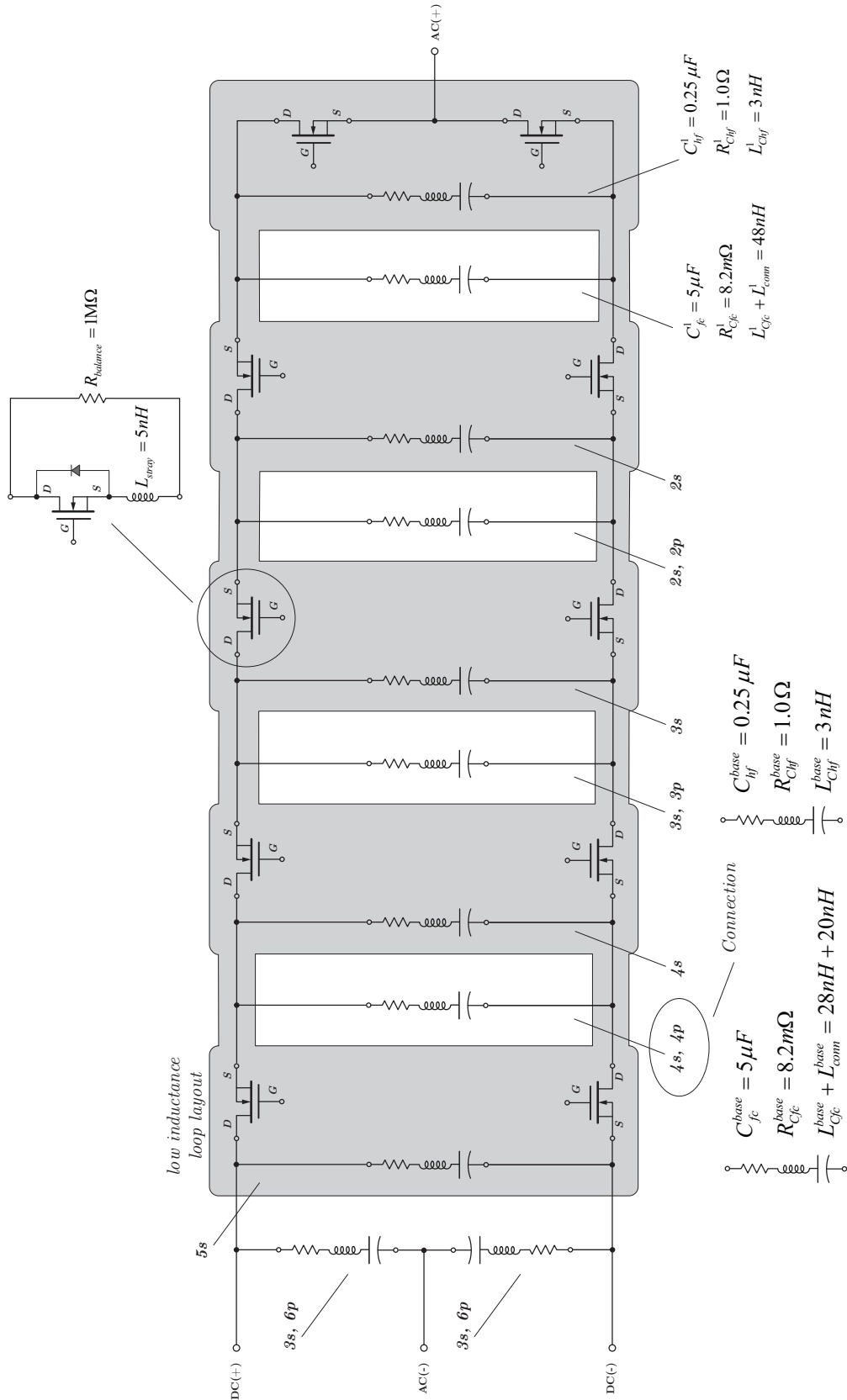
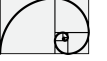
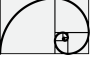


Figure 4: Half-bridge six-level flying capacitor converter - high frequency electrical model.



1.1 System pre-charge

The flying capacitors need to be precharged. A pre-charge control strategy is here proposed (see also Figure 5) as follows

- at time zero the switch s_1 is closed;
- from start of the pre-charge up to the end, a modulation index with negative ramp is applied;
- when modulation index reach the value of zero, the output control voltage starts and the load is connected.

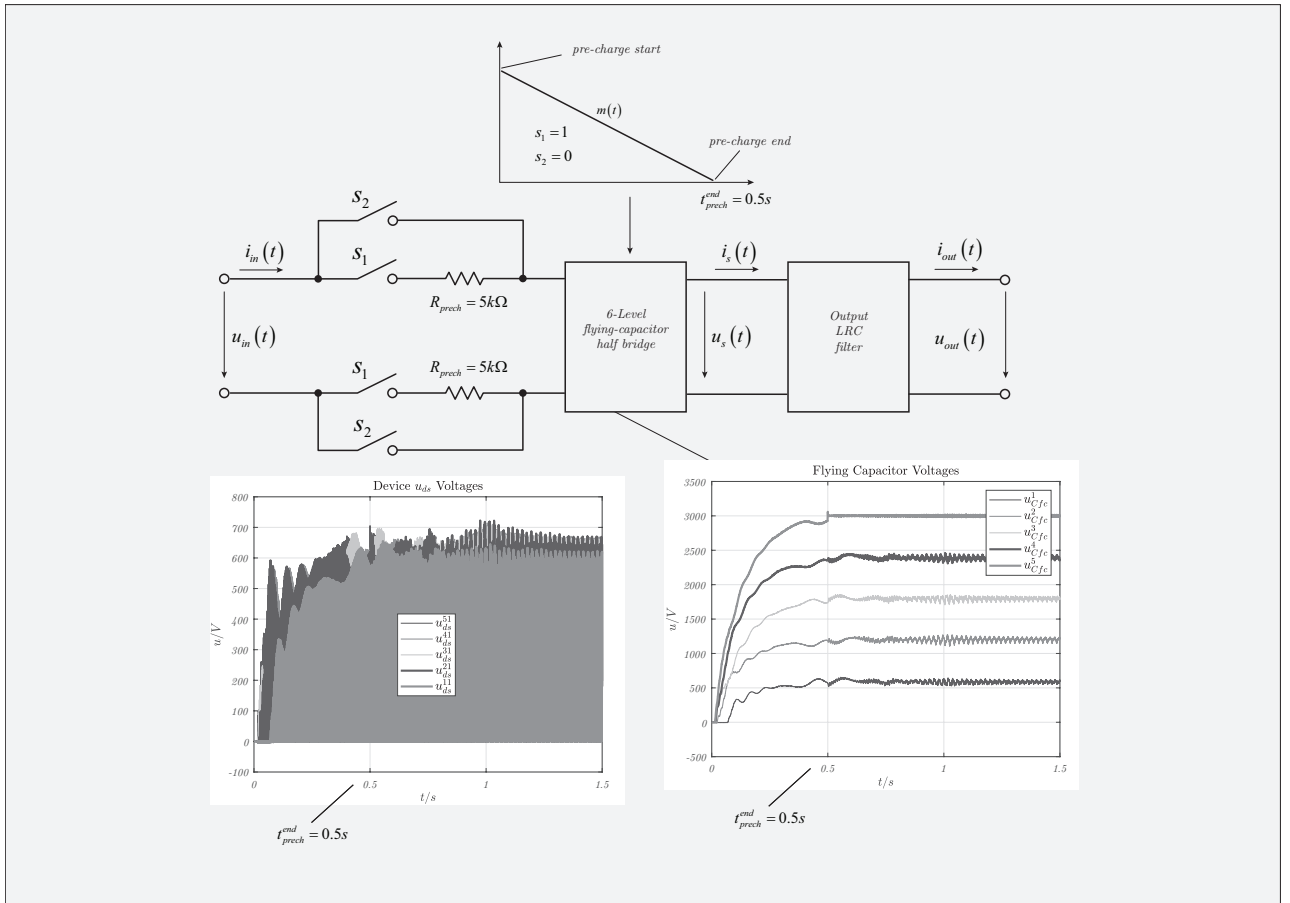
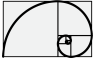


Figure 5: Flying capacitors pre-charge control strategy.

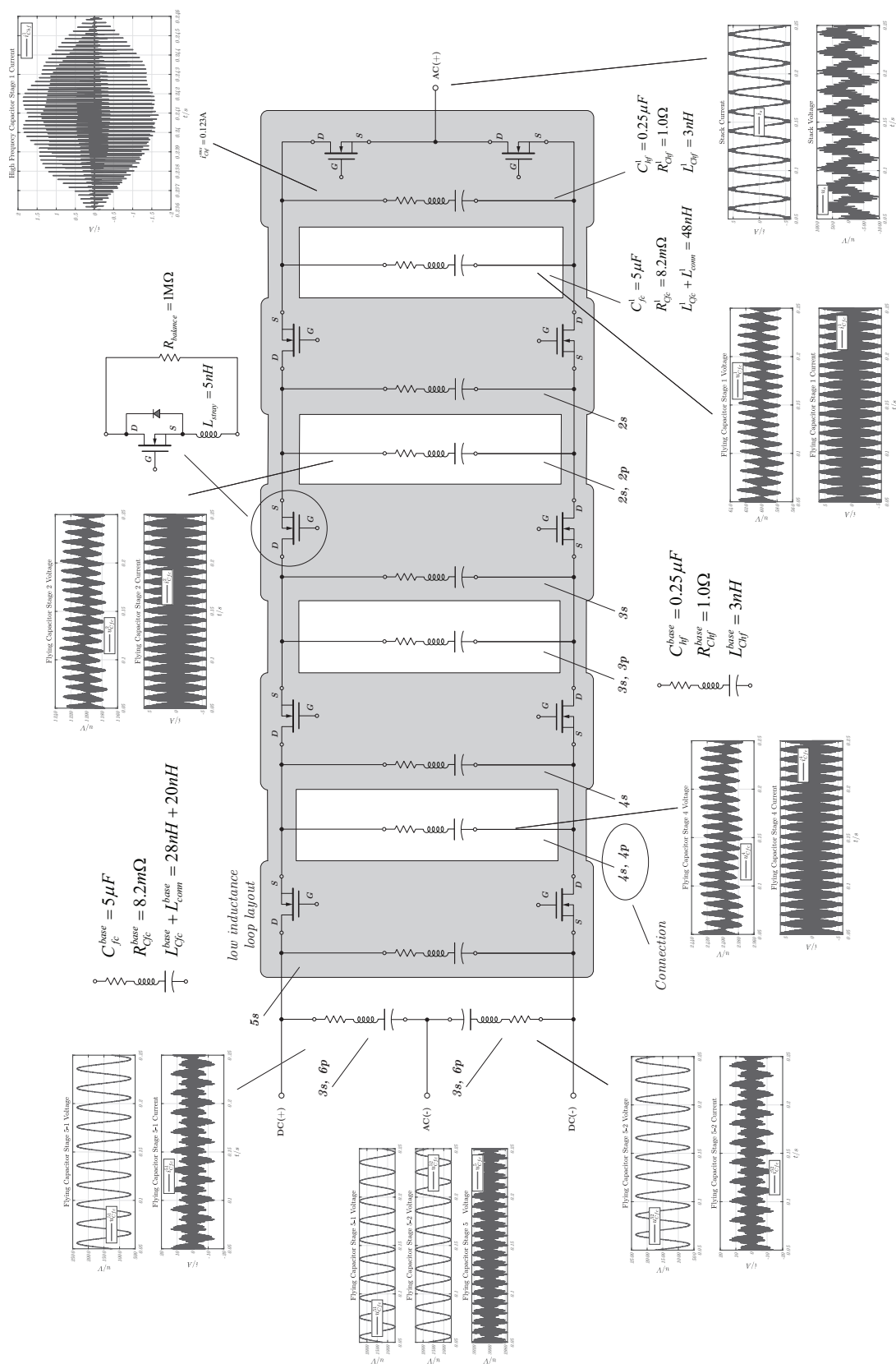
2 High Frequency Analysis

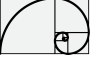
In this section the high frequency simulation results are shown. Basically the whole system has been implemented and simulated using a reasonable modelization of the stray loop inductance of the active and passive components with their wiring.



$$\begin{cases} C_{fc}^{base} = 5 \mu\text{F} \\ R_{Cfc}^{base} = 8.2 \text{ m}\Omega \\ L_{Cfc}^{base} = 28 \text{ nH} \\ L_{connection}^{base} = 20 \text{ nH} \end{cases} \quad (2.1)$$

$$\begin{cases} C_{hf}^{base} = 0.25 \mu\text{F} \\ R_{Chf}^{base} = 1.0 \Omega \\ L_{Chf}^{base} = 3 \text{ nH} \end{cases} \quad (2.2)$$





3 Control Layout

The control implementation of the single phase inverter is basically implemented using the following objects:

- a phase generator at frequency $\omega_0 = 2\pi 50 \text{ rad s}^{-1}$;
- two observers based on $\ddot{\vec{x}}(t) + \omega_0^2 \vec{x}(t) = 0$ plant;
- an outer magnitude voltage control loop;
- an inner vector current control loop, see also Figure 7.

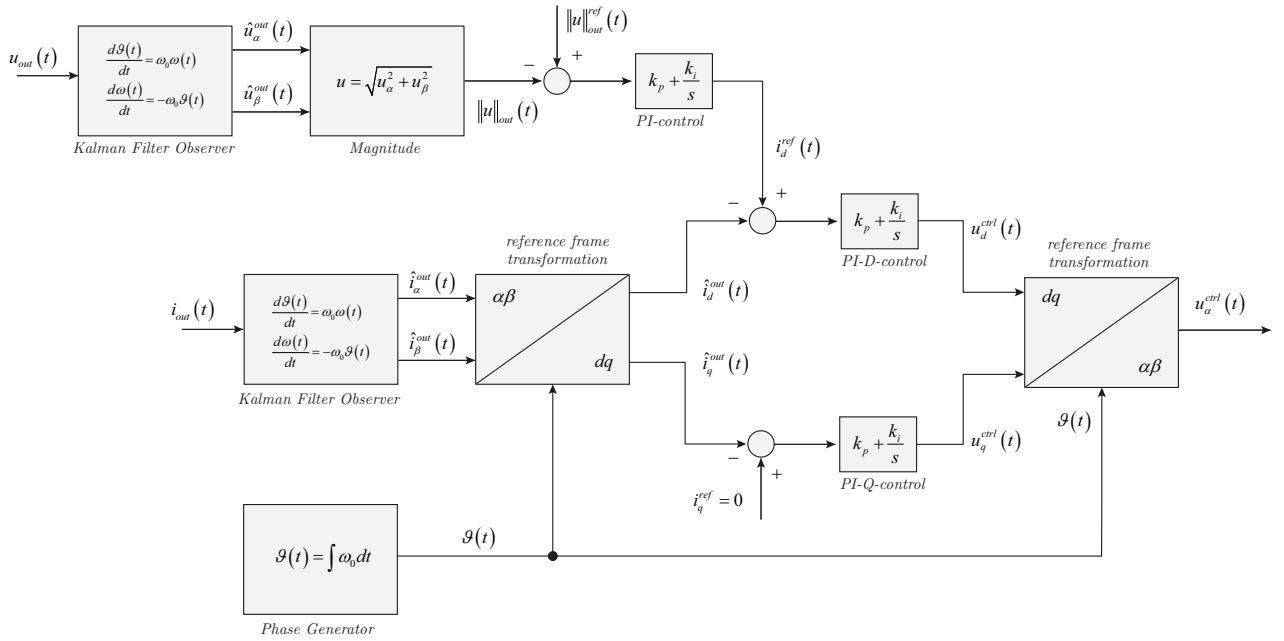


Figure 7: Control Layout.

3.1 Observer based control description

Let's try to consider the voltage and current measures as follows

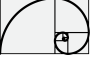
$$\begin{cases} u_{out}(t) = U_{out} \cos(\vartheta) \\ i_{out}(t) = I_{out} \cos(\vartheta - \varphi) \end{cases} \quad (3.1)$$

where the argument $\vartheta(t) = \int \omega_0 dt$.

The core of the control system showed in Figure 7 lays into extrapolation of the quadrature component of both $u_{out}(t)$ and $i_{out}(t)$. From literature exist different solutions, but here the observer based method will be pursued.

The observer model can be assumed as follows

$$\ddot{\vec{x}}(t) + \omega_0^2 \vec{x}(t) = 0 \quad (3.2)$$



where its matrix representation is

$$\begin{cases} \dot{\vec{x}}(t) = \tilde{\mathbf{A}} \vec{x}(t) \\ y(t) = \mathbf{C} \vec{x}(t) \end{cases} \quad (3.3)$$

where

$$\tilde{\mathbf{A}} = \begin{bmatrix} 0 & \omega_0 \\ -\omega_0 & 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (3.4)$$

The equivalent discrete time representation of the system (3.3) can be derived as follows

$$\mathbf{A} = e^{\tilde{\mathbf{A}}t_s} = \mathcal{L}^{-1} \left[\left(s\mathbf{I} - \tilde{\mathbf{A}} \right)^{-1} \right] \Big|_{t=t_s} = \begin{bmatrix} \cos(\omega_0 t_s) & \sin(\omega_0 t_s) \\ -\sin(\omega_0 t_s) & \cos(\omega_0 t_s) \end{bmatrix} \quad (3.5)$$

finally, the equivalent discrete time representation of the system (3.3) become

$$\begin{cases} \vec{x}(k+1) = \mathbf{A} \vec{x}(k) \\ y(k) = \mathbf{C} \vec{x}(k) \end{cases} \quad (3.6)$$

The state observer algorithm used to extract the quadrature component of the measured quantity $y_m(t)$ namely: $u_{out}(t)$ or $i_{out}(t)$ will be based on Kalman filter as follows

State Observer based on Kalman Filter Algorithm

$$\hat{\vec{x}}(k|k) = \mathbf{A} \hat{\vec{x}}(k|k-1) + \mathbf{B} \vec{u}(k|k-1) \quad a \text{ priori} \quad (3.7)$$

$$\mathbf{P}(k|k) = \mathbf{A} \mathbf{P}(k|k-1) \mathbf{A}^T + \mathbf{Q} \quad a \text{ priori} \quad (3.8)$$

$$\mathbf{L}(k) = \mathbf{P}(k|k) \mathbf{C}^T \left(\mathbf{C} \mathbf{P}(k|k) \mathbf{C}^T + \mathbf{R} \right)^{-1} \quad (3.9)$$

$$\hat{\vec{x}}(k+1|k) = \hat{\vec{x}}(k|k) + \mathbf{L}(k) \left(y_m(k) - \mathbf{C} \hat{\vec{x}}(k|k) \right) \quad a \text{ posteriori} \quad (3.10)$$

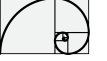
$$\mathbf{P}(k+1|k) = \left(\mathbf{I} - \mathbf{L}(k) \mathbf{C} \right) \mathbf{P}(k|k) \quad a \text{ posteriori} \quad (3.11)$$

where $y_m(t)$ is the measured quantity: $u_{out}(t)$ or $i_{out}(t)$.

Where Q and R have been selected as follows

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & t_s \end{bmatrix}, \quad \mathbf{R} = 1/t_s. \quad (3.12)$$

where $t_s = 200 \mu s$.



3.2 Modulation strategy

The modulation strategy is implemented using the interleaving mode among the levels (see Figure 8). The interleaved modulation strategy activates an kind of self flying capacitor voltage balancing ([1] and [2]), as simulation results shown.

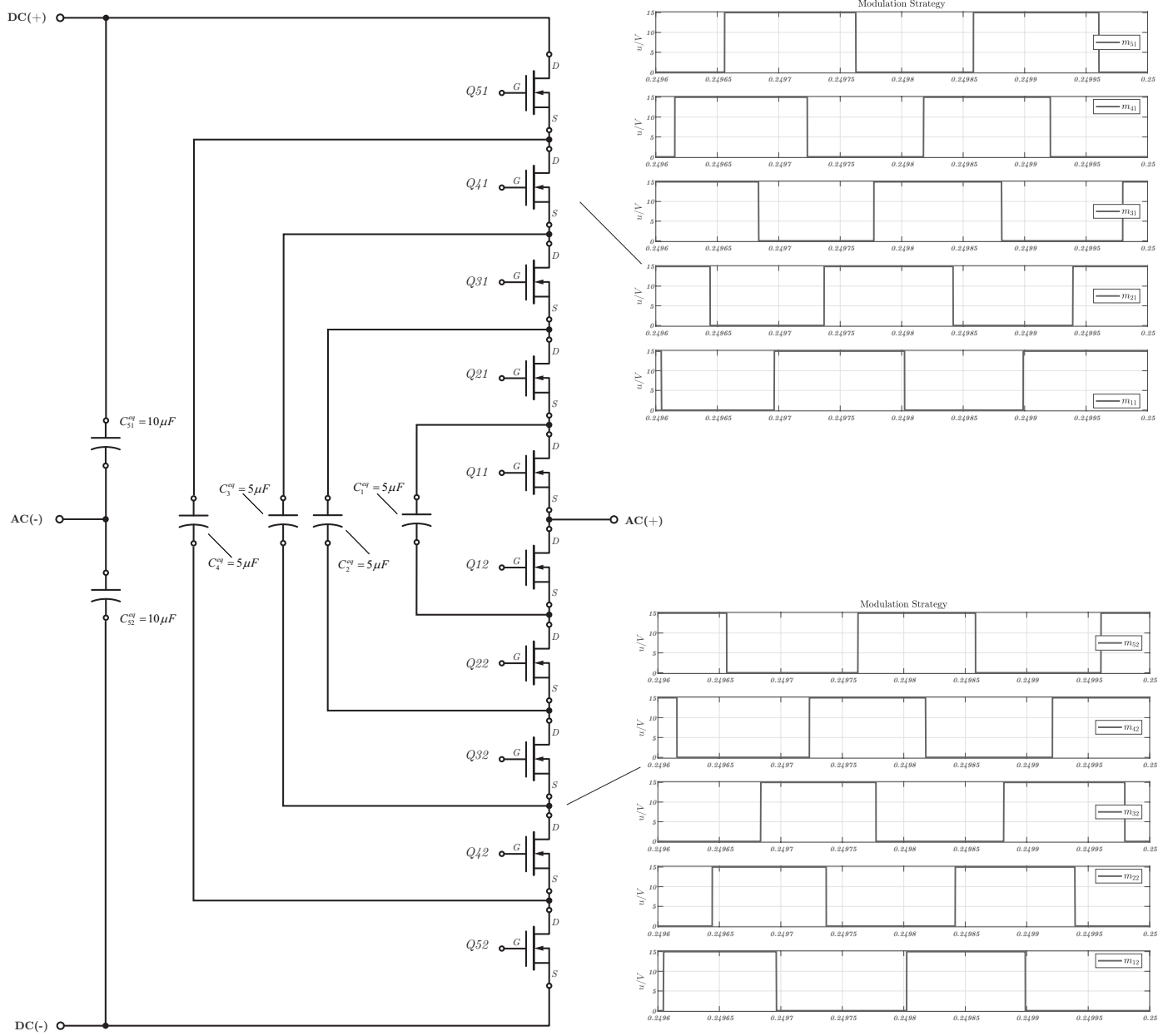


Figure 8: Modulation strategy for self flying capacitor balancing - a different overview.

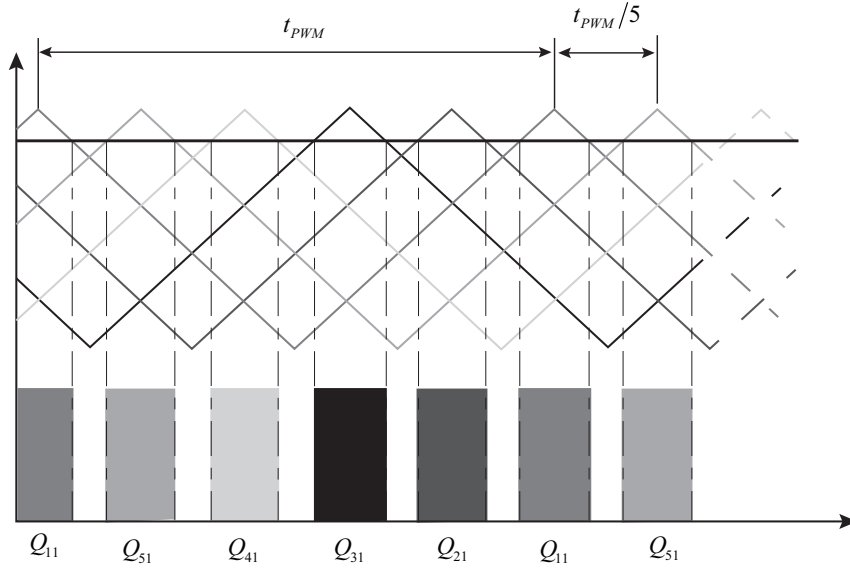
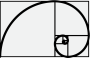


Figure 9: Modulation strategy for self flying capacitor balancing.

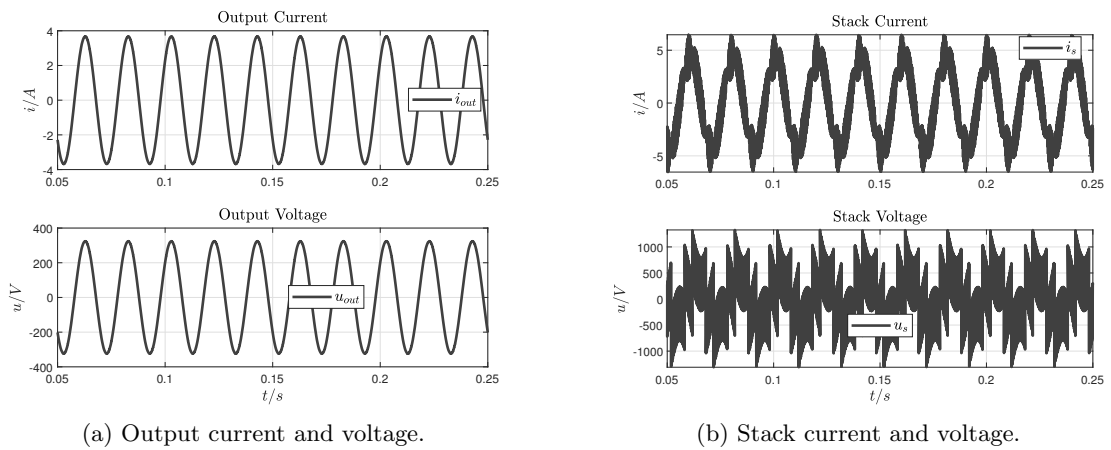
4 Simulation Results

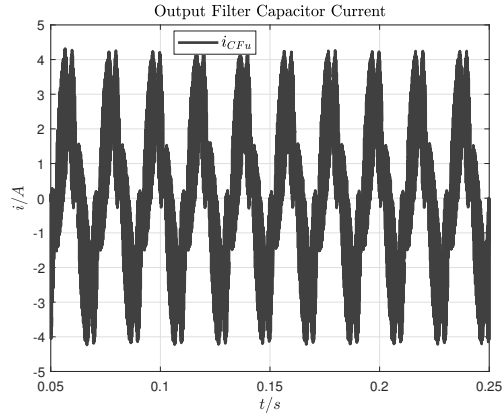
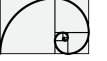
4.1 High frequency open loop control at 5kV of input voltage.

The following set of simulation results have been realized under the following conditions

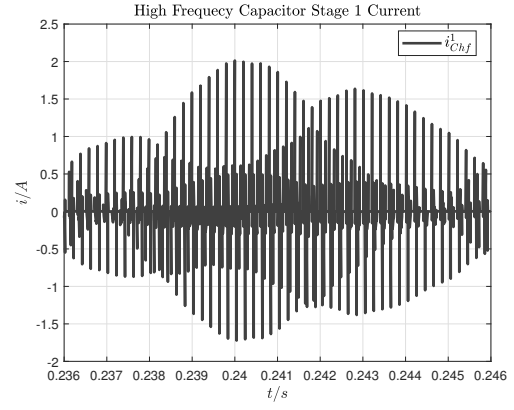
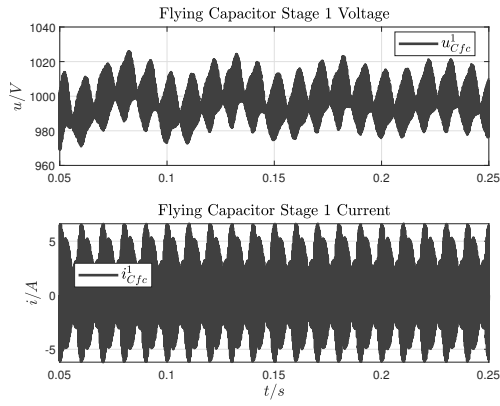
- discrete time step at $t_c = 1 \times 10^{-8}$ s;
- open loop control;
- $u_{dc} = 5$ kV;

and they are intended for electronic layout evaluation.

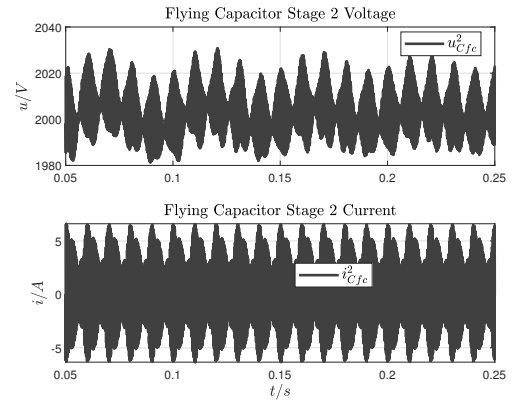
Figure 10: Output current and voltage stages, at $u_{dc} = 5$ kV.



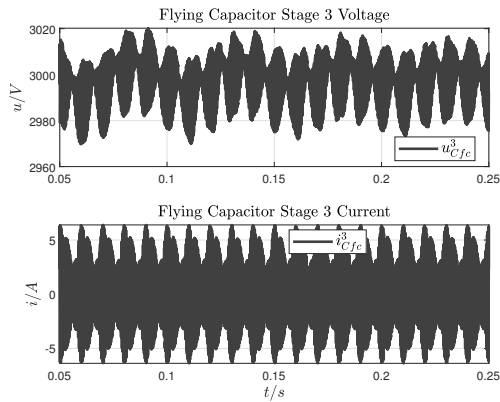
(a) Output filter capacitor current.

(b) High Frequency capacitor C_{hf}^1 current ($i_c^{rms} \approx 0.1$ A).Figure 11: Output filter capacitor current and high frequency dclink current, at $u_{dc} = 5$ kV.

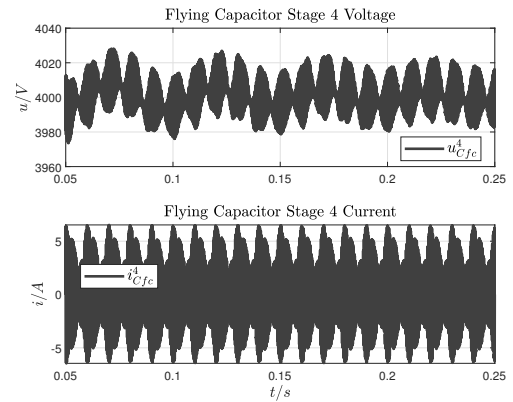
(a) Flying capacitor stage 1: current and voltage.



(b) Flying capacitor stage 2: current and voltage.

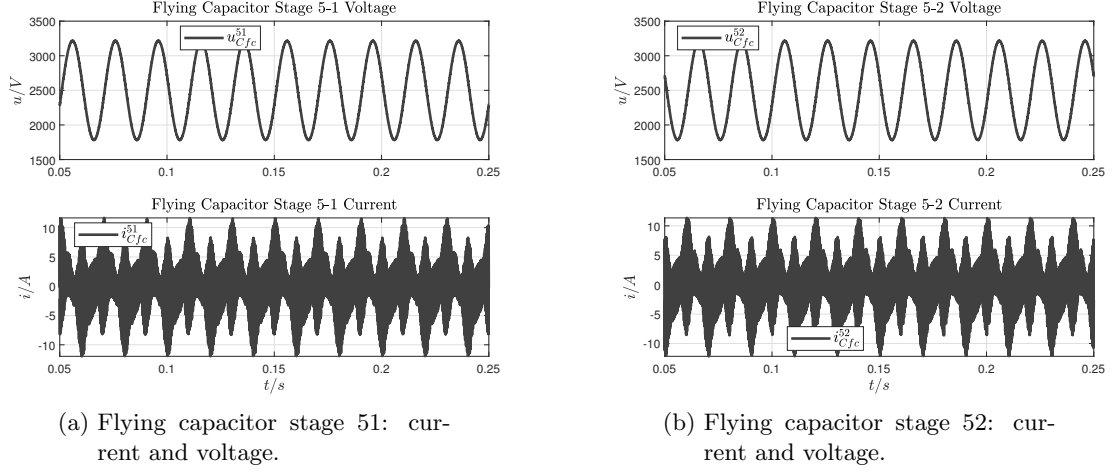
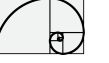
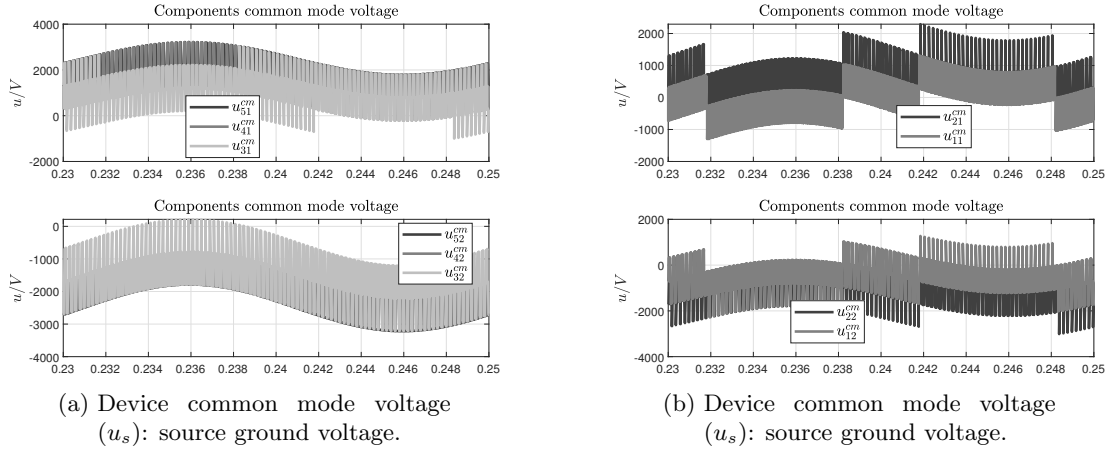
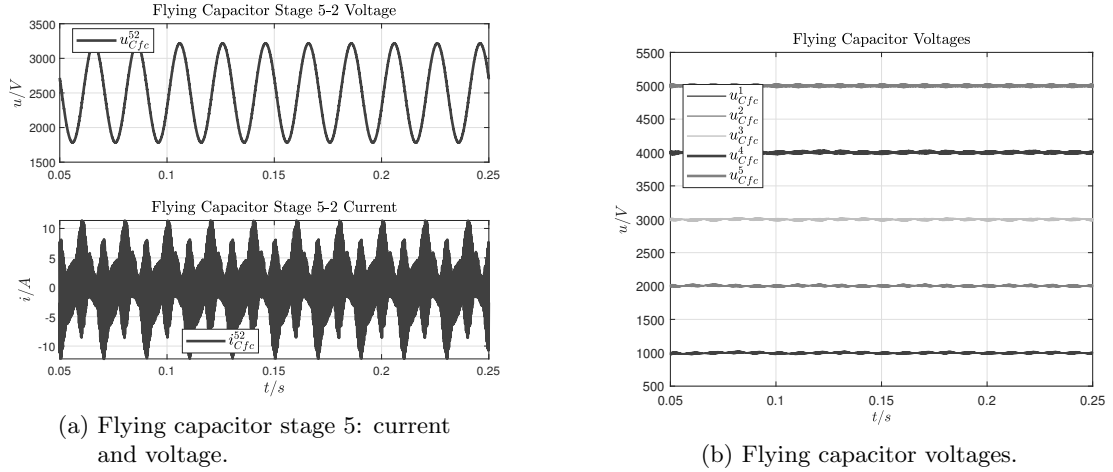
Figure 12: Flying capacitor stages 1 and 2, at $u_{dc} = 5$ kV.

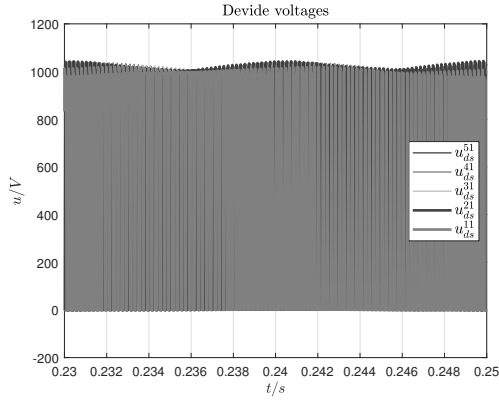
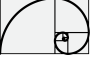
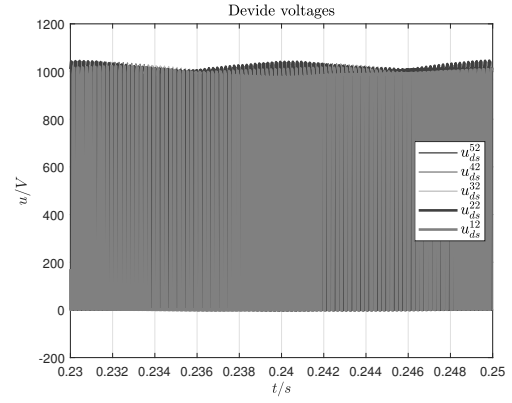
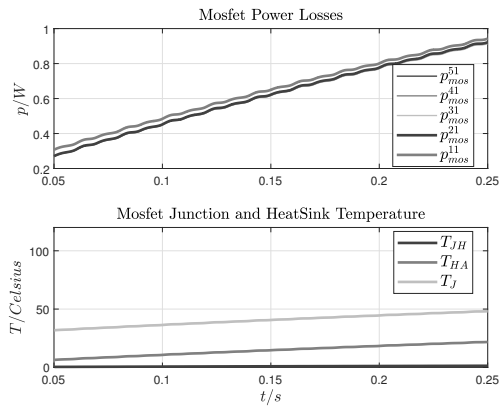
(a) Flying capacitor stage 3: current and voltage.



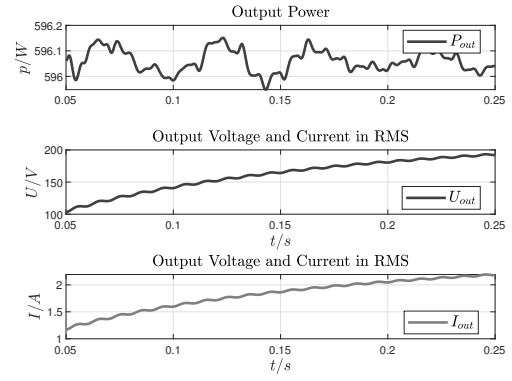
(b) Flying capacitor stage 4: current and voltage.

Figure 13: Flying capacitor stages 3 and 4, at $u_{dc} = 5$ kV.

Figure 14: Flying capacitor stages 51 and 52, at $u_{dc} = 5$ kV.Figure 15: Device common mode voltages, at $u_{dc} = 5$ kV.Figure 16: Flying capacitor voltages, at $u_{dc} = 5$ kV.

(a) Top device voltages (u_{ds}).(b) Bottom device voltages (u_{ds}).Figure 17: Device voltages (u_{ds}), at $u_{dc} = 5$ kV.

(a) Top device power losses.



(b) Active Power and RMS outputs.

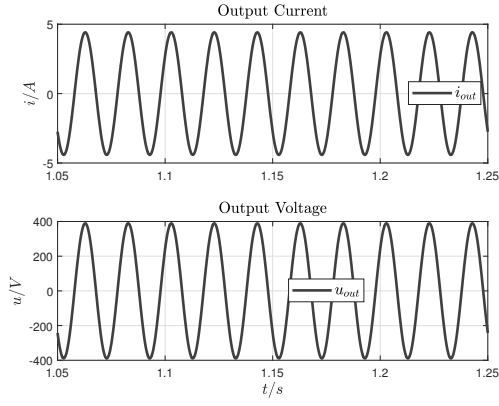
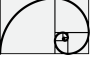
Figure 18: System Performance, at $u_{dc} = 5$ kV.

4.2 High frequency open loop control at 3kV of input voltage

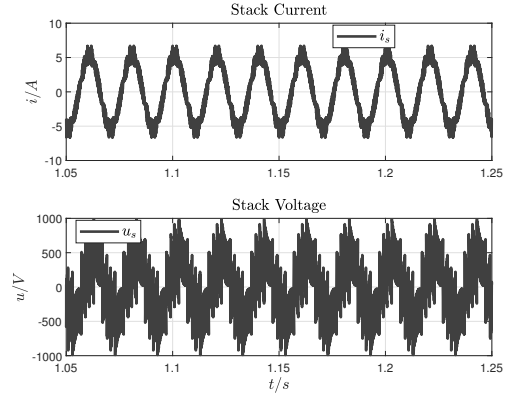
The following set of simulation results have been realized under the following conditions

- discrete time step at $t_c = 1 \times 10^{-8}$ s;
- open loop control;
- $u_{dc} = 3$ kV;

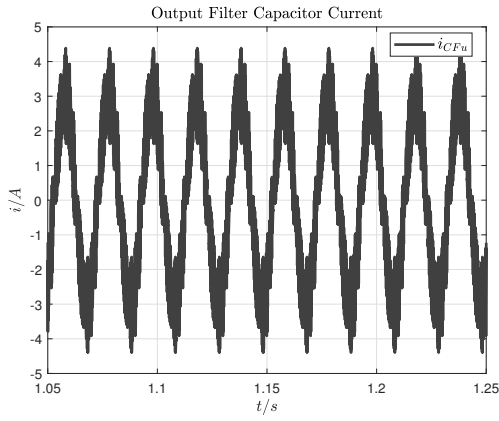
and they are intended for electronic layout evaluation.



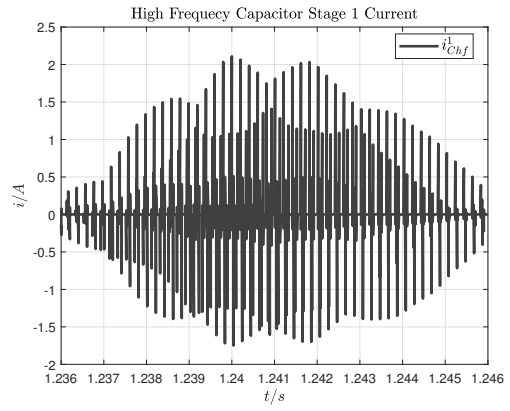
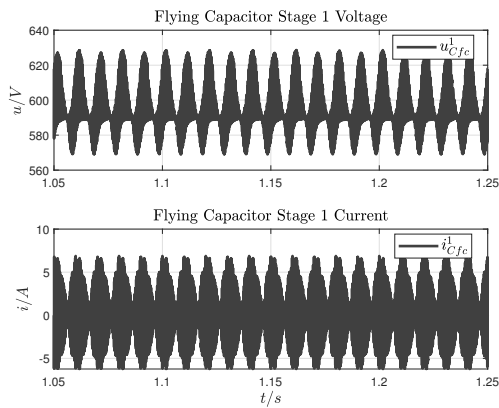
(a) Output current and voltage.



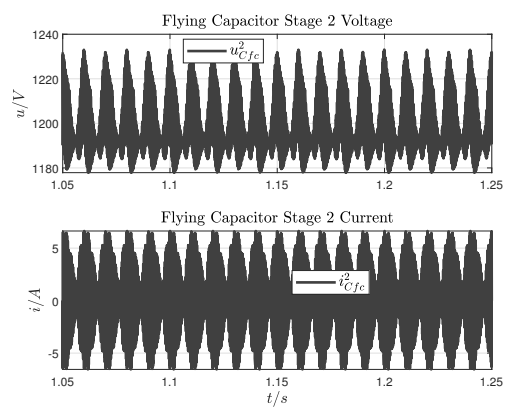
(b) Stack current and voltage.

Figure 19: Output current and voltage stages, at $u_{dc} = 3$ kV.

(a) Output filter capacitor current.

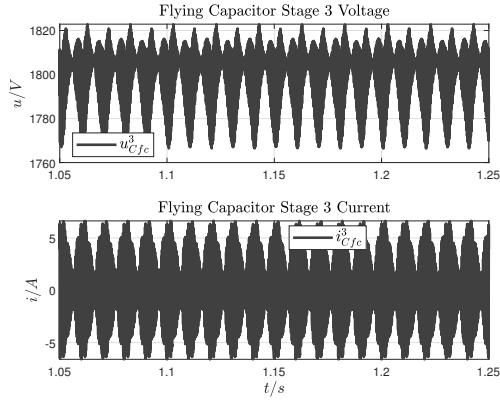
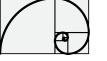
(b) High Frequency capacitor C_{hf}^1 current ($i_c^{rms} \approx 0.1$ A).Figure 20: Output filter capacitor current and high frequency dclink current, at $u_{dc} = 3$ kV.

(a) Flying capacitor stage 1: current and voltage.

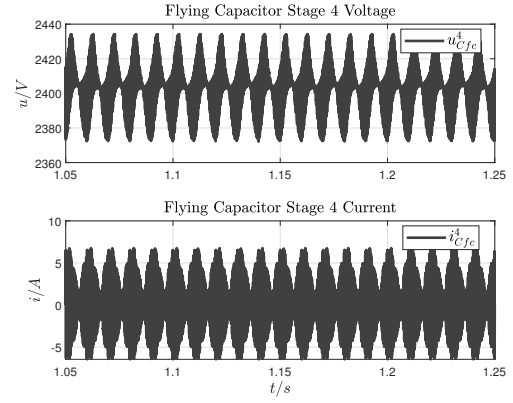


(b) Flying capacitor stage 2: current and voltage.

Figure 21: Flying capacitor stages 1 and 2, at $u_{dc} = 3$ kV.

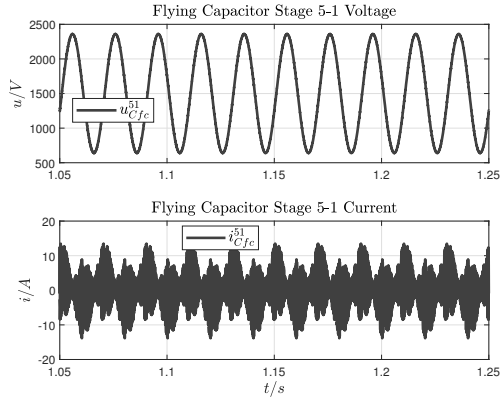


(a) Flying capacitor stage 3: current and voltage.

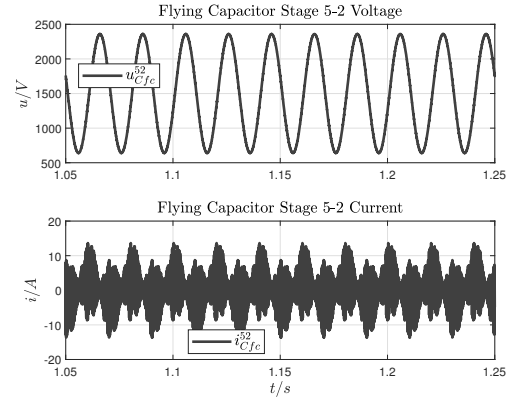


(b) Flying capacitor stage 4: current and voltage.

Figure 22: Flying capacitor stages 3 and 4, at $u_{dc} = 3$ kV.

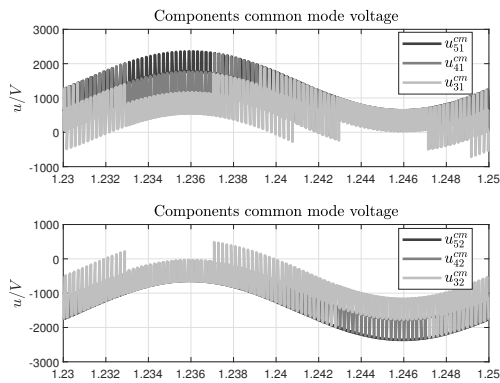


(a) Flying capacitor stage 51: current and voltage.

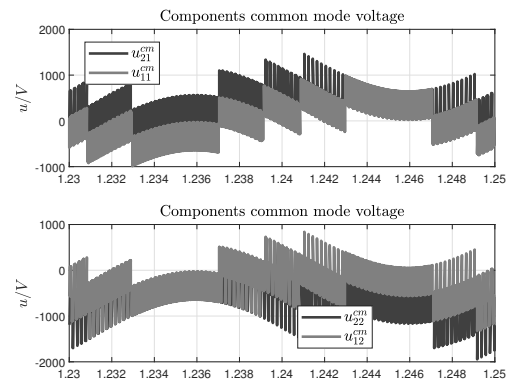


(b) Flying capacitor stage 52: current and voltage.

Figure 23: Flying capacitor stages 51 and 52, at $u_{dc} = 3$ kV.

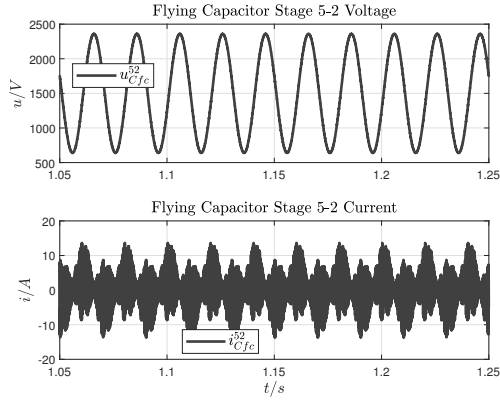
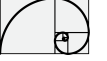


(a) Device common mode voltage (u_s): source ground voltage.

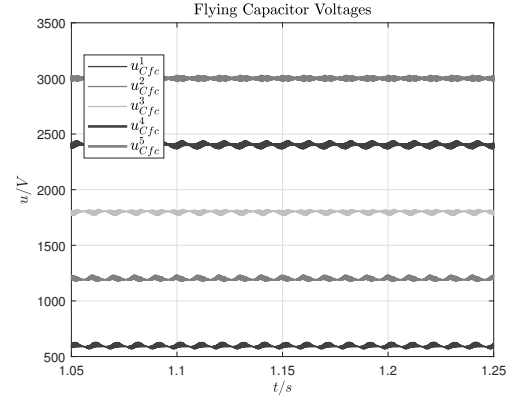


(b) Device common mode voltage (u_s): source ground voltage.

Figure 24: Device common mode voltages, at $u_{dc} = 3$ kV.

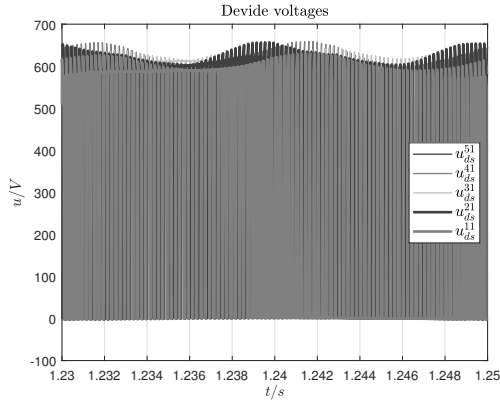


(a) Flying capacitor stage 5: current and voltage.

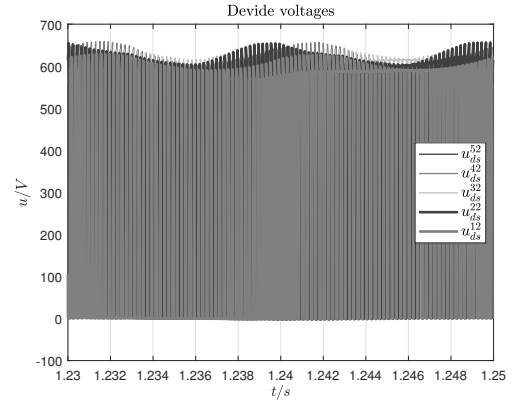


(b) Flying capacitor voltages.

Figure 25: Flying capacitor voltages, at $u_{dc} = 3$ kV.

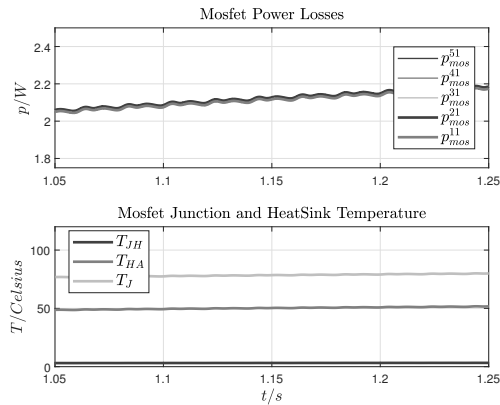


(a) Top device voltages (u_{ds}).

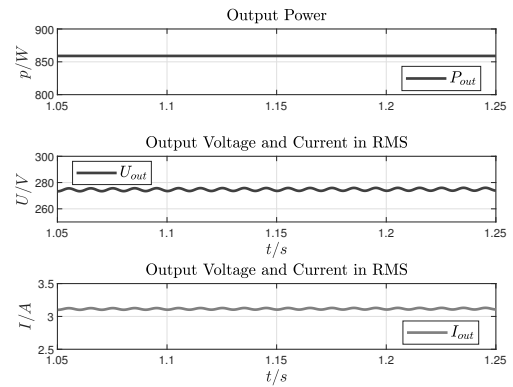


(b) Bottom device voltages (u_{ds}).

Figure 26: Device voltages (u_{ds}), at $u_{dc} = 3$ kV.



(a) Top device power losses.

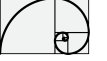


(b) Active Power and RMS outputs.

Figure 27: System Performance, at $u_{dc} = 3$ kV.

4.3 Closed loop control

The following set of simulation results have been realized under the following conditions



- discrete time step at $t_c = 1 \times 10^{-6}$ s;
- closed loop control;
- $u_{dc} = 3$ kV;
- Combination of passive precharge and slope down modulation index is performed from $t_0 = 0$ s to $t_1 = 0.5$ s;
- No load voltage control is performed from $t_1 = 0.5$ s to $t_2 = 1.0$ s;
- At $t_2 = 1.0$ s a 500W step load is connected to the inverter;

and they are intended for control performance evaluation.

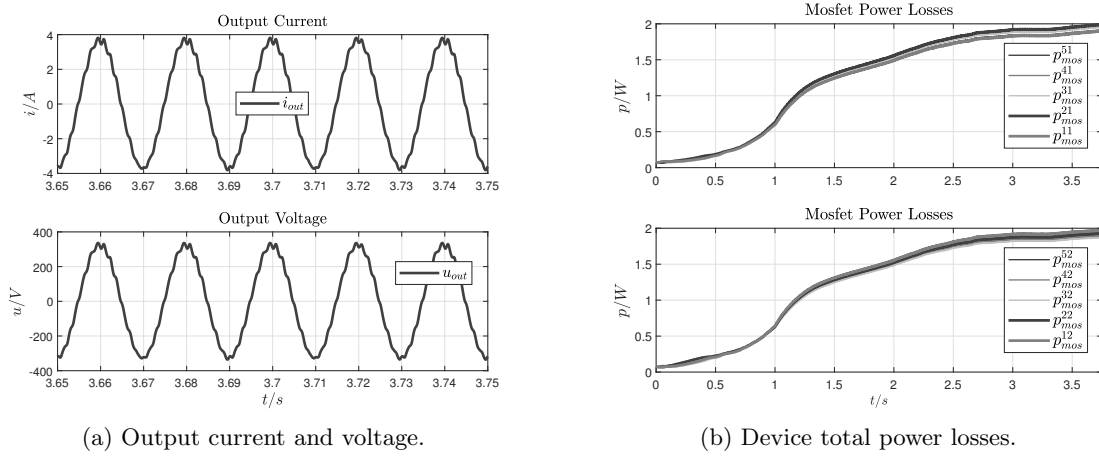


Figure 28: Performance analysis.

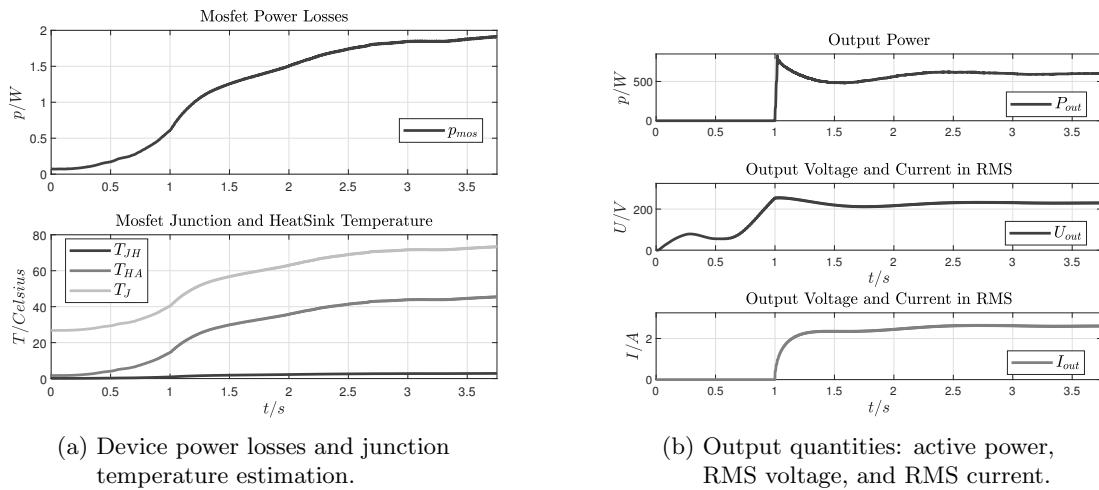


Figure 29: Performance analysis.

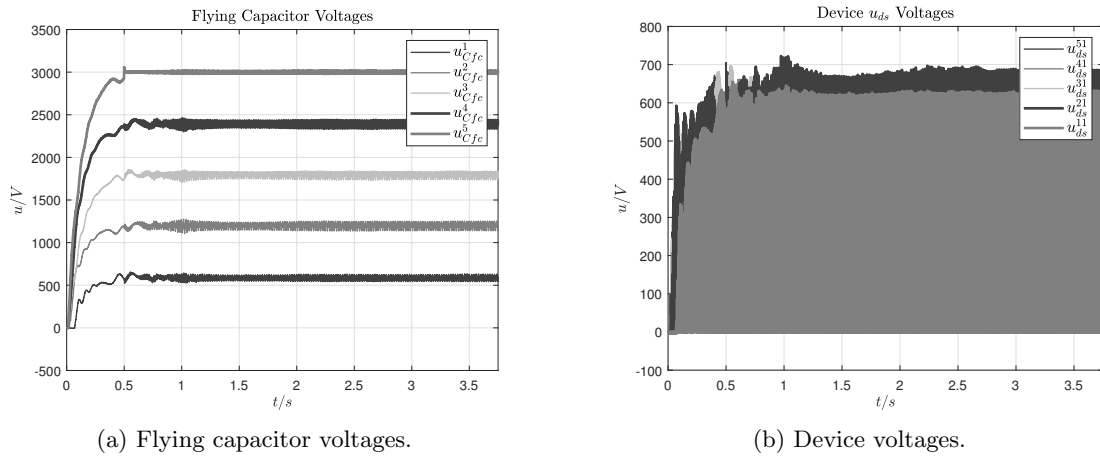
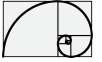


Figure 30: Performance analysis.

References

- [1] R. H. Wilkinson, T. A. Maynard, H. du T. Mouton *Natural Balance of Multicell Converters: The General Case*. IEEE Transaction on Power Electronics, Vol.21, No. 6, November 2006.
- [2] R. H. Wilkinson, T. A. Maynard, H. du T. Mouton *Natural Balance of Multicell Converters: The Two-Cell Case*. IEEE Transaction on Power Electronics, Vol.21, No. 6, November 2006.